and inactivated with phenol at 37°C leaving noresidual live virus.

2. **Beta propiolactone (BPL) Vaccine:** Beta propiolactone is used as the **inactivating agent** instead of Phenol.
3. **Infant Brain Vaccine:** The enceptalitogenic factor in brain tissue is a basic protein associated with myelin.

Vaccines were developed using infant mouse, rat or rabbit brain. Infant brain vaccine is impractical in India.

Non-neural Vaccines

Non-neural vaccines includes

1. Egg Vaccines
2. Tissue Culture Vaccines
3. Subunit Vaccine

Passive Immunisation

Human rabies immune globulin **(HRIG)** is free from the danger of sensitization but should be ensured free from HIV and hepatitis viruses.

Vaccines for Animals

Antirabies immunization in animals is to be done as pre-exposure prophylaxis concentrated cell culture vaccines – inactivated virus gives good protection after a single **Intramuscular** injection. Injections are given at 12 weeks of age and repeated at 1–3 years intervals.

10.6 Human Immuno Deficiency Virus

HIV is known as **Human Immuno Deficiency Virus**, the etiological agent of AIDS, belongs to the lentivirus subgroup of the family **Retroviridae**.

Infobits

Detecting HIV sooner

Fourth generation test helps to detect HIV in blood earlier than previously recommended antibody test. It identifies the viral protein, HIV-1 P24 antigen, which appears in the blood sooner than antibodies.

Source: CDC

Structure

HIV is a spherical **enveloped Virus**, about 90-120min size. The nucleo capsid has an outer icosahedral shell and an inner cone shaped core, enclosing the **ribonucleo proteins**. The genome is diploid, composed of two identical single stranded, **positive sense RNA copies**. When the virus infects a cell, the Viral RNA is transcribed by the **reverse transcriptase enzyme**, first into single stranded DNA and then to double stranded DNA (provirus) which is integrated into the host cell chromosome. The virus coded **envelope proteins** are the projecting knob like spikes which binds to the CD4 receptors on susceptible host cells (Figure 10.8).

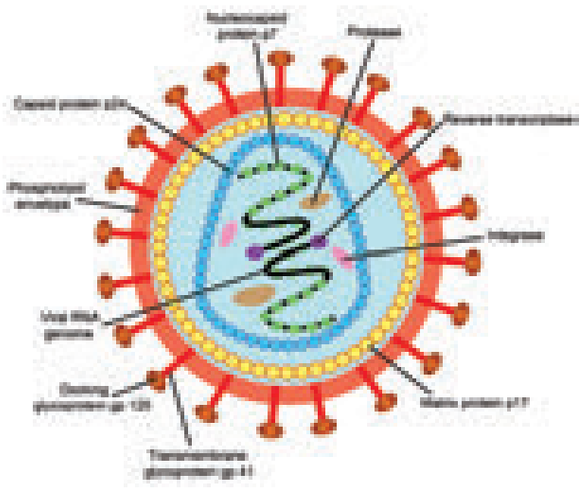


Figure 10.8: Structure of HIV

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